

Project Report: DataCo Supply Chain Analytics

1. Project Overview

This analysis focuses on the DataCo Supply Chain dataset, containing over 180,000 transaction records. The goal is to identify operational inefficiencies—specifically in logistics and delivery reliability—and define the requirements for a monitoring system.

2. Data Classification (Dimensions vs. Measures)

Before processing, the dataset attributes were categorized to determine how data should be grouped (Dimensions) and calculated (Measures).

Dimensions (Qualitative/Categorical)

These answer: Who, Where, When, and What.

- **Time:** Order Date, Shipping Date.
- **Customer:** Segment (Consumer/Corporate), Country, City, State.
- **Product:** Name, Category (e.g., Fishing, Cleats), Department.
- **Logistics:** Shipping Mode (First Class, Standard, etc.), Delivery Status, Order Status.
- **Geography:** Market, Order Region.

Measures (Quantitative/Numerical)

These answer: How Much or How Many.

- **Financials:** Sales, Order Profit, Benefit per Order, Order Item Discount.
 - **Logistics Performance:** Days for Shipping (Real), Days for Shipment (Scheduled).
 - **Volume:** Order Item Quantity, Order Item Total.
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3. Exploratory Data Analysis (EDA) & KPIs

The EDA process was used to uncover hidden patterns within the supply chain.

Key Observations:

1. **The Logistics Paradox:** Many orders are delivered late despite having planned shipping schedules. Specifically, premium "First Class" shipping shows a higher failure rate than "Standard Class."
2. **Profit Concentration:** A small number of product categories (like Fishing) contribute to the majority of the company's total profit.
3. **Regional Disparity:** Certain global markets generate high sales volume but suffer from very low profit margins due to high shipping costs or local delays.

Primary KPIs for Tracking:

- **On-Time Delivery Rate (OTDR):** Percentage of orders delivered by the scheduled date.
 - **Late Delivery Risk:** Probability index of an order being delayed.
 - **Total Sales & Profit:** High-level revenue and net earnings after costs.
 - **Average Shipping Delay:** The gap between actual and promised delivery days.
 - **Profit per Order** – Profit generated from each order
 - **Sales per Customer** – Revenue generated per customer
 - **Late Delivery Rate(%)** - Percentage of Orders which were late
 - **Shipping Mode Performance Comparison** - Comparison of how different shipping methods perform.
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4. System Requirements

4.1 Functional Requirements (What the system must do)

- **FR1: Data Ingestion & Cleaning:** The system must read the CSV files, automatically fill missing values (using mean/mode), and fix date formats for analysis.
- **FR2: Live Dashboard:** A "Scorecard" must display real-time percentages of on-time orders and identify delay-prone regions.
- **FR3: Smart Alerts:** If a shipping method (e.g., First Class) falls below a 10% on-time rate, the system must trigger an automatic warning.
- **FR4: Profit Reporting:** Create visual reports comparing total sales against actual profit to find "money-losing" products.
- **FR5: Role-Based Views:** Logistics staff should see shipping maps, while Management should see financial summaries.

4.2 Non-Functional Requirements (Performance & Security)

- **NFR1: Speed:** The system must refresh charts for 200,000 rows in under 5 seconds.
- **NFR2: Accuracy:** Financial data must be calculated to exactly two decimal places with no rounding errors.

- **NFR3: Scalability:** The backend must be capable of handling a growth to 1 million records without crashing.
- **NFR4: Usability:** The interface must be simple enough that any report is accessible within three clicks.
- **NFR5: Data Privacy:** Personal customer details (Emails/Passwords) must be hidden (masked) from the analytical view.